b. Qualitative conversion to a coloured or light absorbing substance. For example, after de-proteinization of the sample, the supernatant conataining glucose is made to react with orthotoluidine reagent to give a greenish blue coloured complex.

c. Measurement of light absorption of sample. For example, the colour intensities of the complex can be measured.

d. Calculation of the concentration of the substance. Comparing the molecular extinction with that of a relative standard solution of known concentration.

$$\text{Concentration of unknown} = \frac{\text{Absorbance of unknown}}{\text{Absorbance of known (Standard)}} \times \text{Concentration of known}$$

10.6.4 UV absorption spectrophotometry

Absorption spectrophotometry utilizes the phenomena of absorption of light by the sample being analysed. Different materials absorb different wavelengths based on the molecular and chemical nature of the compound. Absorption spectra can use any form of waves. The common type of waves used in absorption spectra include infrared, gamma, atomic, X-ray and visible light. The common wavelength region used in the absorption spectra is .

Applications of spectrophotometry

- Spectrophotometry is used in diverse area in the field of biological science.
- To identify classes of compounds in pure and biological samples.
- For quantitative analysis of proteins, lipids and nucleic acid etc.
- To assay the enzyme activity and their kinetics.
- Elucidation of the structure of organic compounds.
- Measurement of growth kinetics.

10.7 Biosensor

A biosensor is an analytical device which converts a biological response into an electrical signal. The term 'biosensor' is often used to cover sensor devices used in order to determine the concentration of substances and other parameters of biological interest even where they do not utilise a biological system directly. The analyte is coupled to various process including enzyme, antibody and later transduced to obtain a measurable signal. Examples include glucometer and several other diagnostic applications

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